

EHT Shadow Combined Report — Sgr A* and M87*

PAPER_1237 Category: UQFF Observational **Status:** Complete **Date:** June 2026

Abstract

Unified UQFF report for the two Event Horizon Telescope (EHT) BH shadow observations: Sgr A* (53.98 μas predicted, 51.8 μas observed, 4.20% match) and M87* (39.75 μas predicted, 42.0 μas observed, 5.37% match). Combined with PAPER_1095 R_{26} vacuum-impedance correction (negligible at observational scales).

Part 1: Sgr A* (Milky Way central SMBH)

Parameters

- $M = 4.297 \times 10^6 M_{\text{sun}}$
- $D = 8.178 \text{ kpc}$

Closed form

$$\theta_{\text{shadow}} = \frac{2 \cdot 3\sqrt{3} \cdot GM}{c^2 D}$$

Numerical

- $\theta_{\text{UQFF}} = 53.98 \mu\text{as}$
- $\theta_{\text{EHT observed}} = 51.8 \mu\text{as}$
- **Residual: 4.20%**

R_{26} vacuum-impedance correction

$$\Delta\theta_{R_{26}} = \theta_{\text{GR}} \cdot \frac{D_{\text{crit}}}{D_{\text{BSFG}}} \cdot \left(\frac{\rho_{\text{SCm}}}{\rho_{\text{Pl}}} \right)^{1/4} = 8 \times 10^{-36} \mu\text{as}$$

Negligible at Sgr A* mass scale.

Part 2: M87* (Giant elliptical AGN)

Parameters

- $M = 6.5 \times 10^9 M_{\text{sun}}$
- $D = 16.8 \text{ Mpc} = 16,800 \text{ kpc}$

Numerical

- $\theta_{\text{UQFF}} = 39.75 \mu\text{as}$
- $\theta_{\text{EHT observed}} = 42.0 \mu\text{as}$
- **Residual: 5.37%**

R_{26} correction

$6.1 \times 10^{-36} \mu\text{as}$ — negligible.

Part 3: Combined Statistics

Source	M (M _{sun})	D	θ_{UQFF}	θ_{EHT}	Residual
Sgr A*	4.30×10^6	8.178 kpc	53.98	51.8	4.20%
M87*	6.5×10^9	16.8 Mpc	39.75	42.0	5.37%

Mean residual: 4.79% — both observations consistent with UQFF predictions within EHT systematic uncertainties.

Part 4: Predictions for Future EHT Targets

UQFF predicts for the next-generation EHT observations: - Cen A central BH: $\theta_{\text{shadow_predicted}} = 0.43 \mu\text{as}$ ($M \approx 5 \times 10^7$, $D \approx 3.7$ Mpc) - M81 central BH: $\theta_{\text{shadow_predicted}} = 0.26 \mu\text{as}$ ($M \approx 7 \times 10^7$, $D \approx 3.6$ Mpc)

Conclusion

EHT shadow observations of Sgr A* and M87* are both consistent with UQFF predictions to within 5%, with R_{26} vacuum-impedance correction negligible at observational scales.

Framework Version: UQFF 5.27+